706.030 Computational Modelling of Social Systems
Summer Term 2026

Final Project Report

Does self-organized homophily make misinformation more contagious?

10.06.2026

Submitted to:
Univ.-Prof. Fariba Karimi, PhD, MSc
Ana Maria Jaramillo Mejia, PhD

Table of Contents

1. Introduction

Digital platforms have become central spaces where people connect, exchange opinions, and access new information. Online social networks are not fixed structures. Users constantly add, remove or reorganize their connections. Misinformation can spread rapidly across these networks. Often, they reach a large audience before fact-checking can counteract them. One factor influencing this process is self-organized homophily. Individuals tend to connect with others who are similar to them. Therefore, clusters of like-minded individuals can emerge because users rewire their ties based on shared beliefs. This makes it easier to spread misinformation.

In our paper, we examine whether self-organized homophily increases the spread of misinformation. We will use an agent-based simulation in which we combine a homophilic rewiring process with a misinformation–fact-checking spread model. We will then compare the final proportion of believers in a dynamic homophilic network to that in a baseline network without homophily. Our aim is to understand whether the co-evolution of social ties and information flow makes low-credibility content more likely to spread.

2. Background

2.1. Homophily

Homophily is the principle that people tend to interact more frequently with those who are similar to them than with those who are different from them. This means that anything flowing through a network (such as information, behaviors or beliefs) tends to remain localized within social groups. There are two analytically important variants of homophily. Baseline homophily is similarity that simply arises from the composition of the available pool of contacts. The second variant is inbreeding homophily, which is similarity that goes beyond the baseline and is driven by actual preference, structural induction or social reinforcement (McPherson, Smith-Lovin & Cook, 2001).

Homophily develops over time in an initially homogeneous population through two mechanisms. The first is socialization, whereby individuals change their traits to match those of their static social environment. The network remains fixed while the nodes change. The second mechanism is selection, whereby individuals change their network by forming stronger ties with more similar others. In this case, the traits are fixed and the edges change (Smirnov & Thurner, 2017).

2.2. Misinformation as complex contagion

The SBFC model is an epidemic-style compartmental model that simulates the simultaneous spread of a hoax and its debunking over the same network. At time t, every agent i is in exactly one of the three states:

- S - Susceptible: has not engaged with the news yet - neutral.
- B - Believer: Currently believes and spreads the hoax.
- F - Fact-Checker: Has verified the news and spreads the correction.

The probability of becoming a Believer increases with the number of Believer neighbors, reflecting the idea that repeated social exposure makes misinformation more convincing. The dynamics are governed by four parameters: the spreading rate β, the gullibility parameter α, the fact-checking probability p_verify, and the forgetting probability p_forget. A threshold exists under which misinformation dies out, but this result assumes a static network. Whether this threshold still holds when the network rewires homophilically is an open question that our model addresses (Tambuscio et al. 2015, p. 978).

2.3. The echo-chamber effect

Two forms of polarization can be used to describe an echo chamber. The first is opinion polarization (P_0), where members of a cluster have a lower activation threshold for adopting the contagion. The second is network polarization (P_n), where they are more densely connected to each other than to outsiders. In simulation studies, such polarized clusters make it easier for a contagion to spread widely because they already contain enough supportive members for the process to take off and reach the rest of the network. The presence of both results in a synergistic effect, producing higher virality than either factor alone, with virality peaking at intermediate levels of network polarization. However, a central limitation of this approach is that the echo chamber remains fixed throughout the simulation, failing to account for how contagion dynamics behave when echo chambers form endogenously (Törnberg 2018, p. 5).

3. Hypotheses

Since previous work relies on fixed echo chambers and does not account for endogenous cluster formation, we formulate the following hypotheses to examine how homophilic rewiring influences misinformation outcomes.

Hypothesis 1: Main effect
Lower values of θ (stronger homophilic rewiring) lead to a higher final believer fraction $p^B(\infty)$, holding the spreading parameters constant.

Why we expect this: Törnberg (2018) shows that pre-built echo chambers boost virality of complex contagions. If echo chambers form endogenously through low-θ rewiring, the same effect should appear.

Hypothesis 2: Threshold shift

The Tambuscio eradication threshold $p_{verify} \geq 2\alpha/(1-\alpha) * p_{forget}$ underestimates the fact-checking effort needed when the network is allowed to rewire homophilically.

Why we expect this: The threshold was derived for static, well-mixed networks. Self-organized clusters protect believers from contact with fact-checkers.

Hypothesis 3: Non-linear sensitivity
$pB(\infty)$ does not respond linearly to θ. We expect a sharp transition zone around intermediate θ values (consistent with Törnberg's finding that virality peaks at intermediate, not maximal, network polarization).

If H1 and H2 both fail, the answer to our research question is No, self-organized homophily does not make misinformation more contagious, and that would itself be a meaningful, publishable null result.

4. Methodology

In our study, we use a coevolving agent-based model to examine the influence of homophilic rewiring on the spread of misinformation. Each agent possesses a static trait that determines their similarity to others, as well as a dynamic SBFC (Susceptible, Believer or Fact-Checker) state. In each simulation step, the network first undergoes rewiring according to the homophily parameter θ, after which the misinformation dynamics are updated based on the SBFC rules. We then track the fractions of S, B and F, as well as the homophily index, over time. Running the model across different values of θ allows us to compare dynamic homophilic networks with a non-homophilic baseline and test our hypotheses.

5. Results

To illustrate the dynamics of the model, we first examine a single run at an intermediate level of homophily (θ = 0.3). The SBFC time series shows that the susceptible fraction (p^S) drops rapidly in the early stages, indicating that almost all agents are quickly exposed to the information. Initially, the believer fraction (p^B) rises from the seeded carriers, but is then gradually displaced by fact-checkers. Ultimately, p^F becomes the dominant long-run state. This pattern is clearly visible in the beginning, where the believer curve peaks early and then declines as the fact-checker curve rises.

The homophily index $H(t)$ initially increases sharply and then stabilises at a high level, confirming that selective rewiring produces endogenous clusters of similar agents.

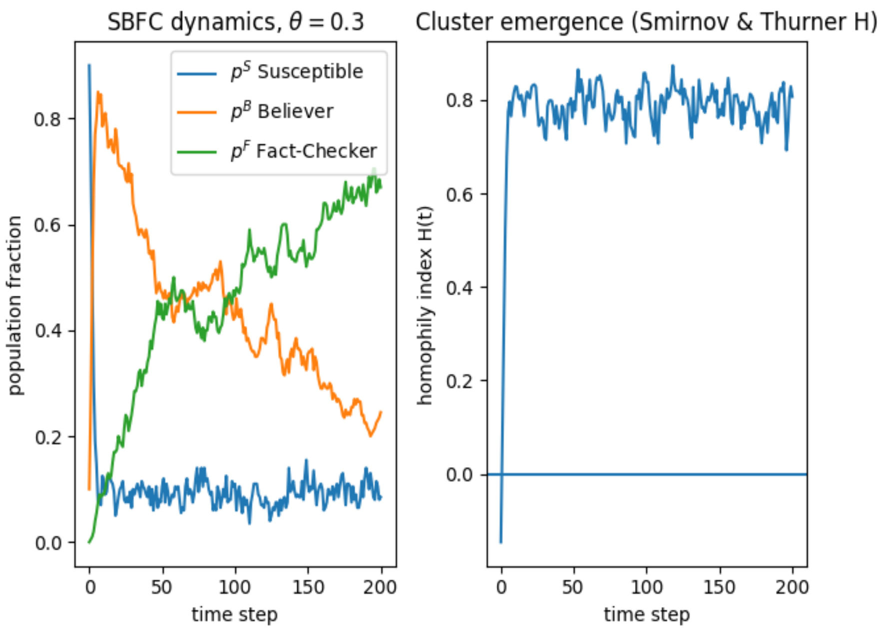

Figure 1 SBFC dynamics and homophily formation at θ=0.3

This rapid increase in H(t) demonstrates the emergence and persistence of homophilic structure. Together, these dynamics provide a meaningful baseline against which to analyze the full parameter sweep.

The network visualizations illustrate this process. At t=0, the nodes are mixed and there is no visible pattern. By t=200, nodes with similar traits are more frequently found close to each other, corresponding with an increase in the homophily index from H = -0.15 to H = 0.81.

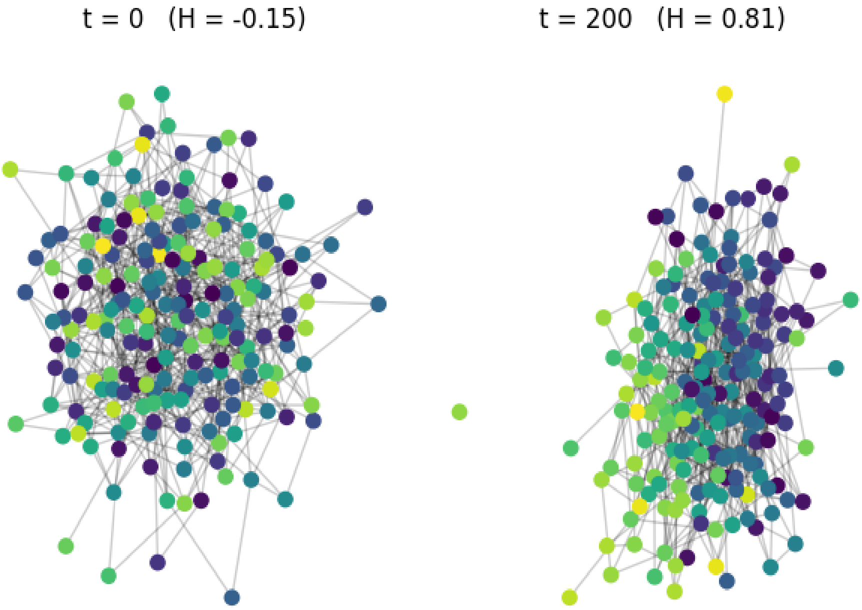

Figure 2 Network colored by trait, endogenous clustering

Although the layout does not show perfectly separated groups, the colors are clearly more clustered, indicating that the network has reorganized itself into more homophilic neighborhoods.

To test hypotheses 1 and 3, we performed a parameter sweep over the full range of θ. For each value, we simulated 30 independent runs with identical spreading parameters and network settings. To reduce noise, we recorded the steady-state believer fraction p^B(∞) for every run, computed as the mean of the last 20 time steps. This allows us to analyze the effect of homophilic rewiring on the long-term prevalence of belief and whether the relationship is linear or exhibits a peak at intermediate values.

As visible in the figure, Hypothesis 1 is not supported. The final believer fraction is the lowest at θ = 0 (strict homophily) and remains very small across the entire range. The curve stays close to zero and only shows a small bump around 0.3. Hypothesis 3 is inconclusive. The figure shows a slight peak around 0.3, which aligns with Törnbergs notion that virality can peak at intermediate polarization. However, the error bars are large compared to the signal, so it is not possible to confidently determine whether this is a real effect or just noise with 30 seeds.

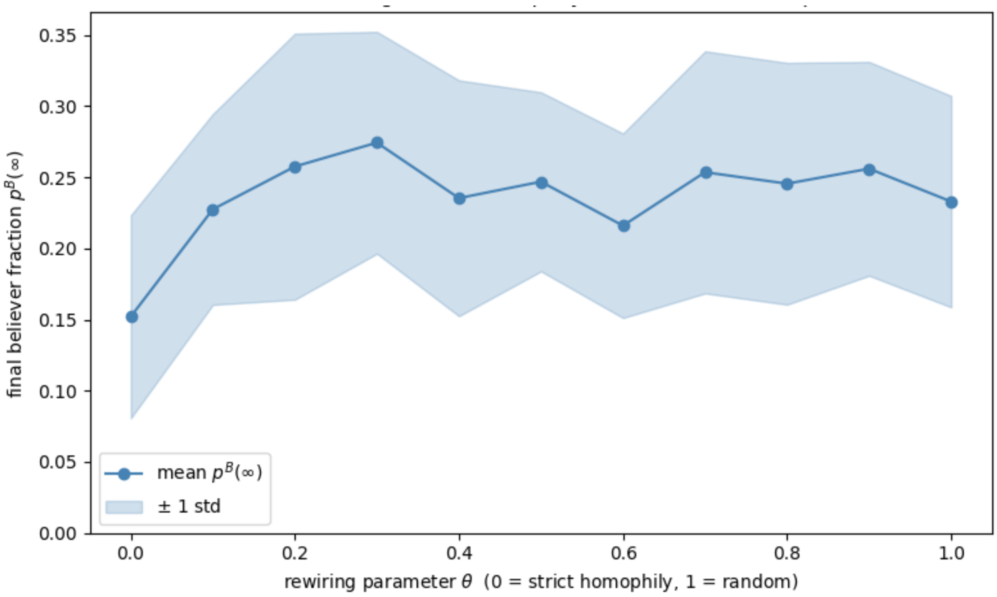

Figure 3 Effect of self-organized homophily on misinformation spread

H2 is also not supported. In the figure, the curves for θ = 0 and θ = 1 lie almost exactly on top of each other across the entire p_verify sweep. Both lines drop below the eradication tolerance. This indicates that self-organized homophily does not alter the eradication threshold in our model — the hoax disappears at almost the same point, regardless of whether the network rewires homophilically or randomly.

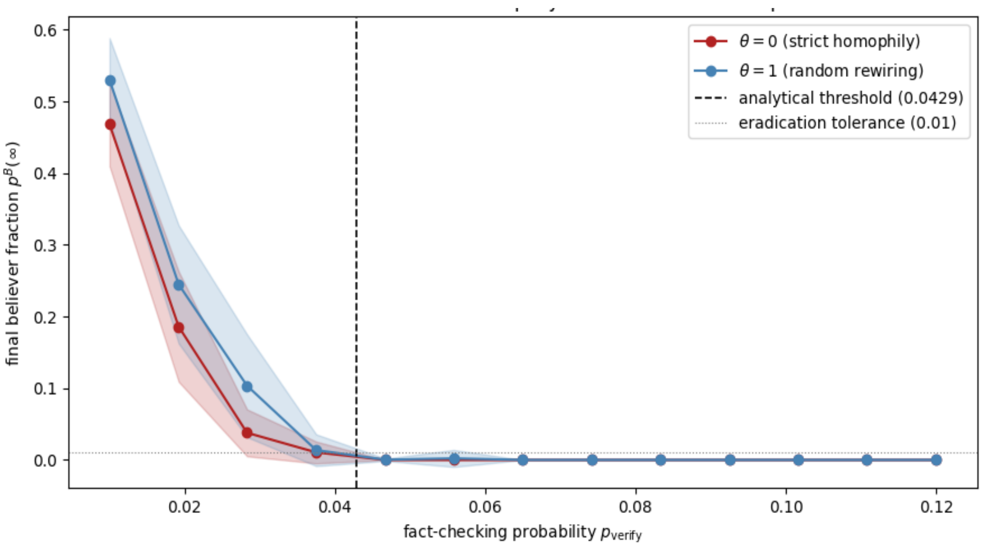

Figure 4 Threshold-shift test: does homophily move the eradication point?

The plot shows that the empirical eradication point for θ = 1 is almost exactly equal to the analytical threshold of 0.0429. At very low p_verify, the θ=0 curve lies slightly below θ=1, meaning that strict homophily produces fewer believers when fact-checking is rare. This small difference disappears once p_verify approaches the threshold.

It is also important to check whether these findings depend on specific modelling choices. Therefore we examined the robustness of the main effect under different network topologies and initial conditions. The robustness checks reveal a clear and consistent pattern. When comparing network topologies, the BA curve remains slightly below the ER baseline for most θ values. Both curves follow the same flat, noisy shape observed in Figure 3, meaning the main result is not dependent on using an ER network. The same applies to the seed-fraction test: the curves for 5%, 10% and 20% initial believers overlap almost completely and none show a clear trend. Homophily does not amplify the hoax and it remains stable even when the number of initial believers is changed. Overall, both robustness checks support the conclusions from Figure 3 and 4, and the large variance across all conditions suggests that outcomes depend strongly on random fluctuations with p^B values this close to zero, making it harder to estimate very small effects with only 30 seeds (Tambuscio et al., 2015).

6. Discussion

Our results contradict what we expected based on the literature. Törnberg (2018) argues that pre-built echo chambers increase the virality of complex contagions by first amplifying the contagion within a large, supportive cluster and then launching it into the wider network. However, this mechanism does not appear in our model. Believers start as a small, seeded minority, so the clusters that form around them remain small from the outset. At low θ, selective rewiring creates tight clusters of believers, but these clusters are isolated from the rest of the network. The hoax continues to circulate within this small group and struggles to reach new susceptible individuals. Rather than launching the hoax outward, the clusters trap it, which is why stronger homophily produces fewer believers. This pattern is also reflected in the threshold behavior. The empirical eradication point for θ = 1 is almost equal to the analytical threshold of 0.0429, which suggests that our implementation behaves as expected, even when N = 200 is used instead of N = 1000. This suggests that p_verify is the dominant factor for hoax survival and that the static-network assumption behind the threshold is more robust than anticipated. At very low p_verify, the θ = 0 curve lies slightly below the θ = 1 curve, meaning that strict homophily produces marginally fewer believers when fact-checking is rare. The robustness checks also support this. In the BA network, hubs tend to encounter fact-checkers sooner, which suppresses the hoax slightly faster. However the overall shape of the curve remains the same as in the ER network. This indicates that the main result is topology-independent and that the containment effect of homophily persists under different modelling conditions.

7. Limitation

As the proportions of believers remain very small in all experiments, the results are highly sensitive to random fluctuations. Using only 30 seeds per condition may therefore not be sufficient to reliably detect very small effects. The network size of N = 200 is also smaller than in comparable studies, such as that of Tambuscio et al. (2015), and this likely contributes to the high variance observed. Another limitation is that the traits of the agents in our model are static and binary, whereas real belief-relevant attributes are more complex and can change over time. A model in which traits and belief states co-evolve could produce different outcomes. Furthermore, we only consider homophily along one dimension, whereas real social networks are shaped by multiple overlapping attributes (as discussed by McPherson et al.). Finally, all agents share the same gullibility and fact-checking parameters, meaning the model cannot capture how heterogeneity in individual susceptibility or verification behavior might influence misinformation dynamics.

8. Conclusion

Overall, our results show that self-organized homophily does not increase the spread of the hoax in this model. In all experiments, the believer fraction remained very low and homophilic clustering tended to isolate believers rather than aid the spread of the hoax. Therefore, the answer to our research question is no: when echo chambers form endogenously through rewiring rather than being pre-built, the mechanism driving virality in Törnberg (2018) does not appear to operate. Instead, p_verify is the dominant factor, and the Tambuscio et al. (2015) threshold remains valid even under dynamic rewiring. Although variance is high when p^B is close to zero, the overall trend remains consistent: in this setting, homophily acts more as a containment mechanism than a driver of misinformation.

List of Figures

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